Ch 6.3, 6.4: PLS; Curse of Dimensionality

Lecture 21 - CMSE 381

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Announcements

Last time:

- PCA/PCR

This lecture:

- Finish PCR
- 6.3: PLS
- 6.4: Issues with higher dimensions

Announcements:

- Homework ~~#5~~ #6 due ~~Friday~~ Mon

Section 1

Last time

Shrinkage

Find β to minimize

$$RSS = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

subject to:

Least Squares:

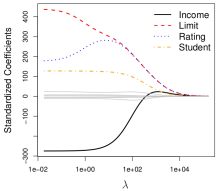
No constraints

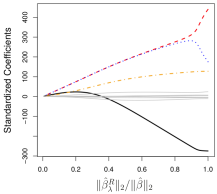
Ridge:

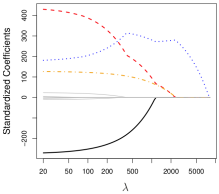
$$\sum_{j=1}^{p} \beta_j^2 \leq s$$

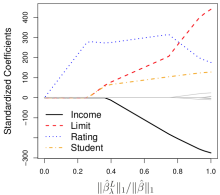
The Lasso:

$$\sum_{j=1}^{p} |\beta_j| \leq s$$









Linear transformation of predictors

Original Predictors:

$$X_1, \cdots, X_p$$

New Predictors:

$$Z_1, \cdots, Z_M \qquad M \ll P$$

$$Z_m = \sum_{j=1}^{p} \varphi_{jm} X_j$$

coef

$$Z_m = \varphi_{1m} X_1 + \varphi_{2m} X_2 + \cdots + \varphi_{pm} X_m$$

The goal:

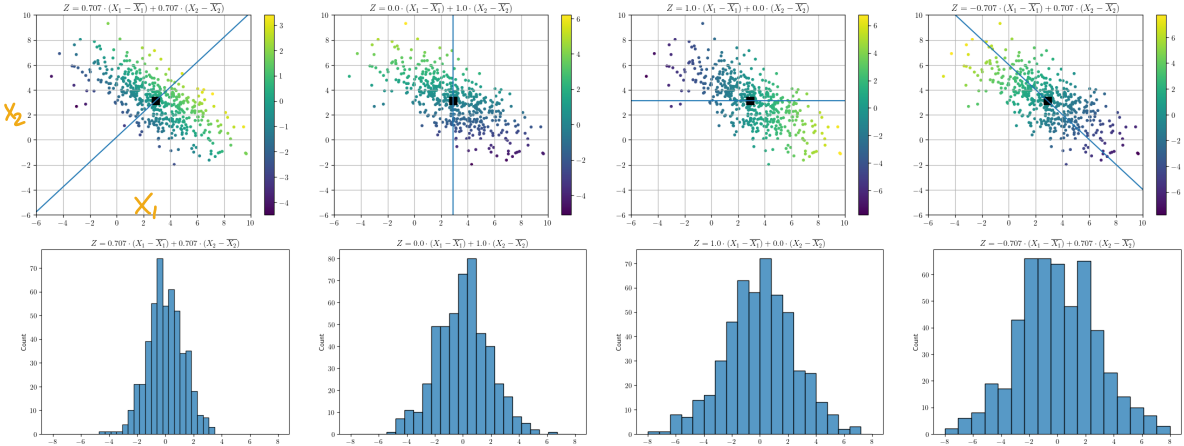
- Find good φ's (PCA)
- Fit regression model on Z_i's using least squares (PLS)

$$y_i = \theta_0 + \sum_{m=1}^{M} \theta_m z_{im} + \varepsilon_i$$

$$= \theta_0 + \theta_1 Z_1 + \theta_2 Z_2 + \cdots \theta_M Z_M$$

- Hope that lower dimensions means less overfitting

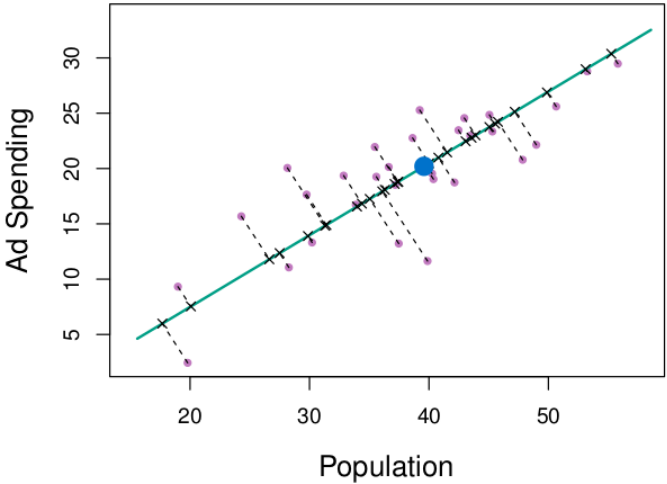
PCA - First PC



low variance

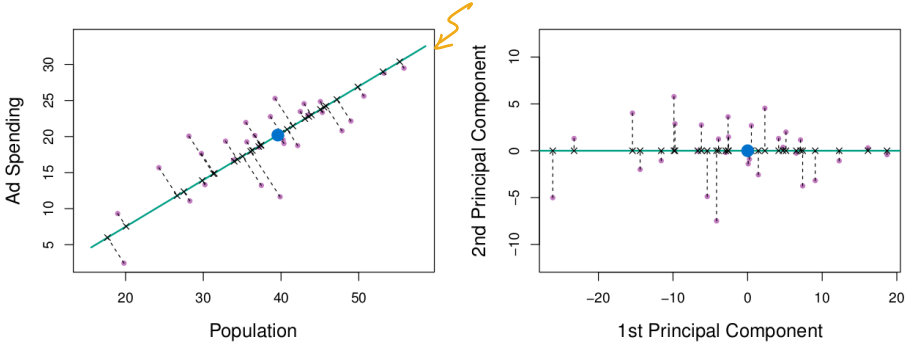
Goal: find line with highest variance

Projection onto first PC



$$Z_1 = 0.839 \cdot (\mathtt{pop} - \overline{\mathtt{pop}}) + 0.544 \cdot (\mathtt{ad} - \overline{\mathtt{ad}})$$

Drawing points in PC space



Input: $X_1, \ldots, X_p$

Can have $Z_1, \ldots, Z_p$

1st PC

p^{th} PC

Interpretation of PCR coefficients

Original Predictors:

$$X_1, \cdots, X_p$$

New Predictors:

$$Z_1, \cdots, Z_M$$

$$Z_m = \sum_{j=1}^{p} \varphi_{jm} X_j$$

Learned model:

$$y = \theta_0 + \theta_1 Z_1 + \cdots + \theta_M Z_M$$

Cannot use θ_i's to say anything about X_i's

$p=3 \quad M=2 \quad \varphi = \begin{pmatrix} 1 & -1 \\ 3 & -1/2 \\ 0 & -1/2 \end{pmatrix}$

$Z_1 = X_1 + 3X_2 + 0 \cdot X_3$

$Z_2 = -X_1 - 1/2 X_2 - 1/2 X_3$

* This is not a selection method! *

Learned: $y = 3 + 2Z_1 - Z_2$

$y = 3 + 2(X_1 + 3X_2 + 0X_3) - (-X_1 - 1/2 X_2 - 1/2 X_3)$

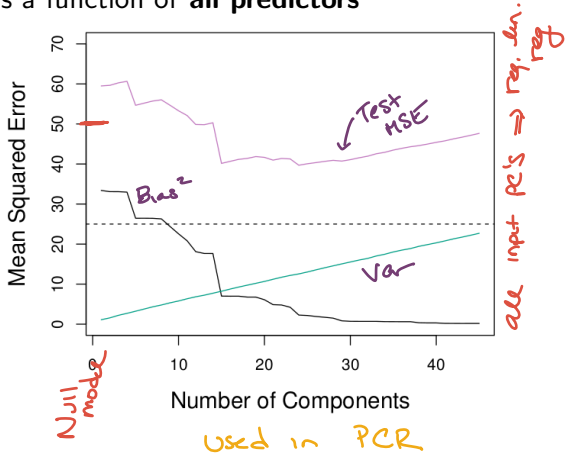
$= 3 + 3X_1 + 2.5X_2 - 0.5X_3$

* Can go back from the θ's to figure out a learned model in terms of X_i's (these would be $\hat{\beta}_i$)

Bias variance trade-off

Example with simulated data: $n = 50$ observations of $p = 45$ predictors

Y is a function of **all predictors**



- U-shape to find best # of components

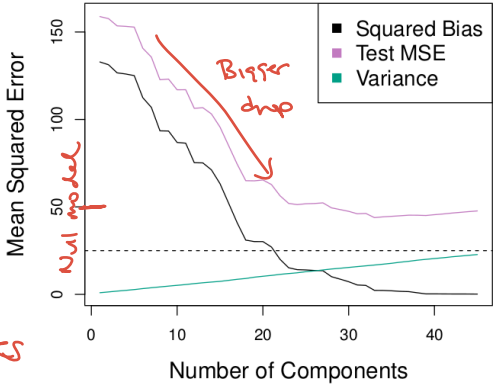
Bias variance trade-off

Example with simulated data: $n = 50$ observations of $p = 45$ predictors
Y is a function of **2 predictors**

↳ expect to do better with fewer inputs

* U-shape

* PCA w/ good choice of # of components is improvement over plain least squares



Properties of PCR

* It does better when info need to predict y is contained in the first few PC's

* # of PC's to use is done w/ k-fold CV
* Standardize data (St dev = 1) otherwise large magnitude inputs skew the PC choices

* Not a feature selection method:
  always have a model using all inputs $X_1, \ldots, X_p$
  even if fewer $Z_1, \ldots, Z_M$'s used.

Section 2

Partial Least Squares (PLS)

Supervised alternative

PCR: Non-supervised

↳ PCA doesn't use y to make decisions about best projections

No guarantee that directions that have highest variance are used to predict Y

Partial Least Squares (PLS):

- Input $x_1, \ldots, x_p$, y
- Identify new features $z_1, \ldots, z_M$

  linear combos of $x_1, \ldots, x_p$

  ↳ find φ's

  But — quality measure involves Y
- Fit linear model using least squares on $z_1, \ldots, z_M$

$$y = \theta_0 + \theta_1 z_1 + \ldots + \theta_M z_M$$

First direction Z_1 for Partial Least Squares (PLS)

Use pop to predict sales

$Sales = \beta_0 + \beta_1 pop$

$\varphi_{pop,1}$

Use ad to predict Sales

$Sales = \beta_0 + \beta_1 ad$

$\varphi_{ad,1}$

- Set φ_{j1} equal to the coefficient from simple linear regression of Y onto X_j
- The first direction is

$$Z_1 = \sum_{j=1}^{p} \varphi_{j1} X_j$$

$$Z_1 = \varphi_{pop,1} X_{pop} + \varphi_{ad,1} X_{ad}$$

Ex. Prediction of Y =Sales (not shown) on X_1 =Population and X_2 =Ad Spending

- Solid green: First PLS direction
- Dashed: First PC direction

Second (and more) PLS directions

- Regress each variable on Z_1 and take residuals
- Compute Z_2 using *orthogonalized data* same as for Z_1

The leftover info not explained by Z_1

prep direction but now uses info from y

* Can choose M (# of input variables) using k-fold CV

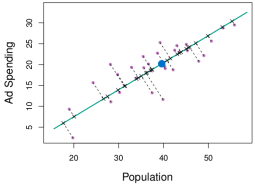
* also not a selection method.

Code example on hitters data

PCR vs PLS

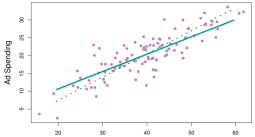
PCR

- Unsupervised dimensionality reduction + linear regression ≈ PCA
- Choose component Z_1 in the direction of most variance using only X_i's information
- Choose Z_2 and beyond by the same method after "getting rid" of info in the directions already explained



PLS

- Supervised dimensionality reduction
- Choose component Z_1 by using simple regression coefficients of each X_i onto Y
- Choose Z_2 and beyond by the same method after "getting rid" of info in the directions already explained
- Not a particular benefit, so usually default to PCA unless you have a good reason for this



Section 3

Issues in Higher Dimensions

High-Dimensional Data

About relationship b/t n and p, not just large p!

Low-Dimensions

$n \gg p$

- Either p is small or small relative to n
- Everything we've done works well here

Ex) simple lin reg: $p = 1$ $n = 100$

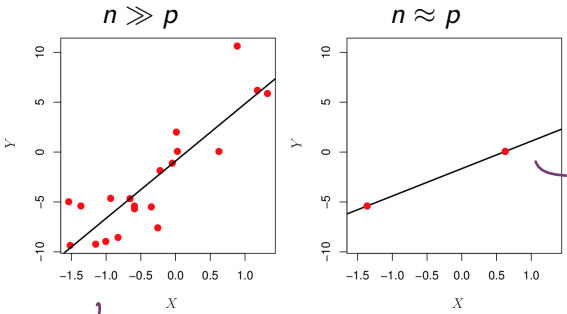
High-Dimensions

$n \ll p$

- B/c of new tech, this is how much more common
- Eg predict blood pressure based on genetic data
  $n \approx 200$ $p \approx 500{,}000$
- In gen, classical approaches fail here
- Issues show up even if p is slightly smaller than n

What goes wrong?

$n \approx 20$ $p = 1$ $n = 2$ $p = 1$



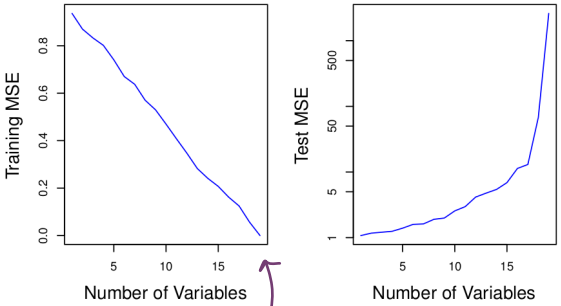
always have perfect fit (because only one possible line)

reasonable fit for y

overfitting ....

More issues with least squares on big p

test score gets high



training goes to 0

Garbage input data

- $n = 20$
- Regression on $p = 1, \cdots, 20$
- Y completely unrelated to variables

The answer to dealing with big p

Be less flexible!

Too many input variables

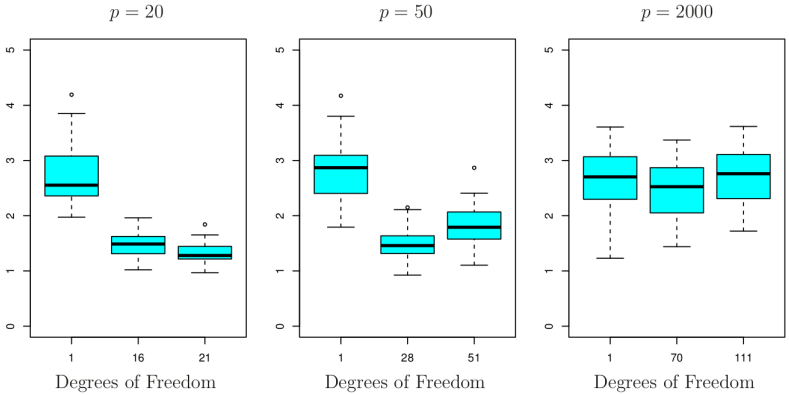
Use less!

Selection methods
- Forward/ Bkwd Best subset
- Lasso

Shrinkage
- Lasso
- Ridge

dim red
- PCR, PLS

Example with Lasso



- $n = 100$
- Boxplots = Test MSE
- DF = # non-zero coeffs

Key points

(1) Regularization or shrinkage plays a key role in high-dim problems

(2) Parameter tuning done w/ K-fold CV

(3) The test error tends to increase with dimensionality of the problem unless the features are truly related to the response.

Curse of dimensionality

Phenomena that arise when analyzing and organizing data in high-dimensional spaces that do not occur in low-dimensional settings.

- Counter-intuitive to "think more variables means better fit"
- Addition of useful features (actually related to response) can improve fit
- But ... adding non-related variables just adds noise
- Increased risk of overfitting

Interpretation in high dimensions

Multi-collinearity: the concept that the variables in a regression might be correlated with each other

High dim: any predictive variable
can be written as
lin combo other predictive variables

Not sure which are actually predictive
↳ many possible models

Reporting errors in high dimensions

↳ Test error!

K-fold CV

Next time

Lec #	Date			Reading	Homeworks
20	Fri	Oct 27	Dimension Reduction	6.3	
21	Mon	Oct 30	More dimension reduction; High dimensions	6.4	
22	Wed	Nov 1	Polynomial & Step Functions	7.1,7.2	
23	Fri	Nov 3	Step Functions	7.2	~~HW #6 Due~~
24	Mon	Nov 6	Basis functions, Regression Splines	7.3,7.4	Hw #6
25	Wed	Nov 8	Decision Trees	8.1	
26	Fri	Nov 10	Random Forests	8.2.1, 8.2.2	
27	Mon	Nov 13	Maximal Margin Classifier	9.1	
28	Wed	Nov 15	SVC	9.2	
29	Fri	Nov 17	SVM	9.3, 9.4	
30	Mon	Nov 20	Single layer NN	10.1	
31	Wed	Nov 22	Overflow/project day?		
	Fri	Nov 24	No class - Thanksgiving		
	Mon	Nov 27	**Review**		
	Wed	Nov 29	**Midterm #3**		
32	Fri	Dec 1	Multi Layer NN	10.2	
33	Mon	Dec 4	CNN	10.3	
34	Wed	Dec 6	Unsupervised Learning & Clustering	12.1, 12.4	
35	Fri	Dec 8	Overflow/Project day?		Project due